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RAGS TO RICHES

The jaws of ragworms may yield a valuable new material

When it comes to looking for advanced engineering materials, the animal kingdom rarely comes to mind. Yet engineers sometimes find that the natural world produces more impressive substances than those devised by the human brain. Spider-silk, for example, is stronger than steel, and is now finding its way into bulletproof jackets. And the ridges and furrows of a gecko's ceiling-grasping toes have inspired a glue-less adhesive tape.

The newest candidate for translation from the animal to human world, though, looks even more unlikely. Dr Chris Broomell and Dr Herbert Waite, of the University of California, Santa Barbara, have been studying the jaws of ragworms – which, as careless fishermen who have used them for bait will attest, can give a nasty nip. NEREIS VIRENS, also known as the sandworm or ragworm, is a burrowing marine worm found in shallow waters in the North Atlantic region. Dr Broomell and Dr Waite were curious about the composition of the only hard parts of an otherwise squishy animal. They were keen to discover what created the remarkable toughness of its jaws, which rivals that of human teeth and exceeds the hardness of many synthetic plastics. In finding out, they may have blundered across the starting point for a new biological material.

In the 1980s, ecologists looking for organisms that could be used as indicators of oceanic pollution took a keen interest in ragworms. These ecologists found that the worms' jaws contained a lot of zinc, a metal that is toxic in large doses. Their initial suggestion was that the worms were dumping excess zinc into their jaws, where it could do no harm, as a way of keeping their bodies free of poison. Measuring the zinc in ragworm jaws was thus thought of as a way of monitoring zinc pollution.

That suggestion, however, fell apart when it became clear that the jaws of worms from clean water, too, were stuffed with similar quantities of zinc. At this point, the ecologists lost interest. But engineers were intrigued. Eventually, Dr Broomell and Dr Waite decided to have a closer look at what was really going on.

They collected the jaws of 1,000 worms and first checked the strength and durability of these by pressing them with a microscopic diamond probe. This revealed that the material from which they are made is as strong as aluminium and impressively light. Most strong biological structures of this sort, such as shells, are highly mineralised. That is, they incorporate crystals of insoluble inorganic salts, most often calcium carbonate or calcium phosphate, in a matrix of protein. This creates a composite material akin to glass-fibre or carbon-fibre. These composites are of interest to engineers in their own right, but when Dr Broomell and Dr Waite stuck ragworm jaws in an X-ray spectroscope they found no sign of mineralisation at all.

That confused them, and they have spent several years trying to find out what is really going on. What they have discovered, as they report in the journal **BIOMACROMOLECULES**, is that the zinc is far from being a pollutant. In fact, it is crucial to ragworm survival. Anyone who has seen them will know that they crawl around beaches and mudflats using tiny structures along the sides of their bodies that work like legs, but are in fact gills. At the front of their bulbous blue heads they have curved fangs that they use to capture and tear apart their crustaceous prey.

In fact, they found that ragworm jaws are made of a mixture of protein and zinc ions. The protein in question contains a lot of an amino acid called histidine. Indeed, it has ten times more of this amino acid than the average protein. Histidine likes to bind tightly to zinc ions. The consequence is that a material composed of histidine-rich proteins and zinc has enormous strength. But, lacking the dense calcium salts of mineralised biological structures, it is also quite light.

These qualities are a desirable combination in an engineering material – and particularly so in those materials used in aircraft. Dr Broomell and Dr Waite have thus passed their discovery on to a group of scientists at NASA, America's aerospace agency, in order that they can try to take the next steps. These are to see whether worm-jaw protein, or something similar, can be made in large quantities, and to try forming it into useful shapes. If both of those prove possible, then an intriguing alternative to traditional composite materials may become available – and worms, in a sense, will fly.